

Yiyang Chen

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Education

Washington University in St. Louis

St. Louis, MO, USA

Ph.D. in *Imaging Science*

2021 - present

M.S. in *Electrical Engineering* (en route to Ph.D.)

2025

- **Dissertation:** Multi-dimensional Single Molecule Nanoscopy to Elucidate the Structure and Conformational Dynamics of Cell Membrane Proteins
- **Advisor:** Matthew D. Lew

Nankai University

Tianjin, China

B.S. (Honors) in *Physics*, Poling Class of Physics

2021

- **Thesis:** The Study of Deformability of Human Erythrocyte Based on Microfluidics
- **Advisor:** Leiting Pan

Publications

*equal contribution

Peer-Reviewed Articles

- [3] **Y. Chen***, Y. Qiu*, M. D. Lew. Single Fluorogens and Orientation-Localization Microscopy for Quantifying Chemical and Biomolecular Dynamics at the Nanoscale. *Acc. Chem. Res.* **59**, 2693-2708 (2026).
- [2] E. Gillett, S. Chatterjee*, J. Chatterjee*, N. Kovalenko, C. Xu, D. Fan, **Y. Chen**, Y. Qiu, J. Miao, V. Nelavoy, M. D. Lew, M. P. Backlund, C. F. Landes. Fused deep learning enables 6D single-molecule localization in polarization-resolved microscopy. *Methods Appl. Fluoresc.* **14**, 035006 (2026).
- [1] **Y. Chen**, Y. Qiu, M. D. Lew. Resolving the Orientations of and Angular Separation Between a Pair of Dipole Emitters. *Phys. Rev. Lett.* **134**, 093805 (2025).

Manuscripts in Preparation

- [1] X. Li*, **Y. Chen*** et al. DeepSMOLM3D: a deep learning approach for sequential estimation of 3D localization and 3D orientation in single-molecule microscopy.

Conference Proceedings

- [1] **Y. Chen**, Y. Qiu, M. D. Lew. Combining Excitation and Emission Modulation Resolves the Angular Separation between a Pair of Dipole Emitters. In *Optica Biophotonics Congress 2025*, Technical Digest Series (Optica Publishing Group, 2025), paper NM1C.4.

Research Experiences

Single-Molecule Orientation-Localization Microscopy: Fundamental Limits, Computational Methods, and Biophysical Applications

2022 – present

Graduate Research Assistant | Advisor: Matthew D. Lew | Preston M. Green Department of Electrical & Systems Engineering, Washington University in St. Louis

Theoretical Modeling & Analysis

- Discovered the fundamental degeneracy and limits for distinguishing a pair of spatially overlapping dipole emitters from rotating single molecules; derived the mathematical proof of the necessity of measuring fourth-order orientation moments.
- Developed a combined excitation polarization and emission dipole-spread function (DSF) modulation to improve the precision of angular separation measurements by 200–400% and centroid orientation by 50% over state-of-the-art methods.
- Investigating fourth-order-moment measurements to extend the measurable parameter space for higher-order orientational dynamics.

Computational Imaging & Algorithm Development

- Engineered DSFs by maximizing Fisher information using convex optimization approaches for targeted orientation regimes.
- Designed a physics-informed SMOLM data generation pipeline to create high-fidelity training and validation datasets for DeepSMOLM3D, ensuring accurate modeling of complex 6D imaging parameters.
- Conducted comprehensive simulation studies (varying SNR, molecule density, etc.) and experimental trials on cell membrane imaging to demonstrate superior accuracy, precision, and reconstruction speed over traditional

maximum-likelihood estimation under challenging imaging scenarios.

Optical Systems & Biophysical Applications

- Executed full optical alignment and calibration of a dual-polarization 4f imaging system incorporating a Liquid Crystal Spatial Light Modulator (SLM) for precise phase modulation.
- Performed PAINT single-molecule imaging of supported lipid bilayers (SLB) and cell membranes using membrane-sensitive fluorogenic fluorophores, demonstrating different binding dynamics of different fluorophores on membrane structures.
- Developing protein-tag-based labeling strategies for measuring membrane-protein orientation and rotational dynamics in cells.
- Conducted hydroxyphenyl fluorescein (HPF) single-molecule imaging to observe and quantify redox reactions in biomolecular condensates. Discovered interface-dependent redox reactions in G3BP stress granule assembly factor 1 (G3BP1) condensates.

Study of Human Red Blood Cell (hRBC) Membrane Characteristics Based on Microfluidics and Single-Molecule Localization Microscopy

2019 – 2021

Undergraduate Research Assistant | Advisor: Leiting Pan | School of Physics, Nankai University

- Quantified hRBC deformability by measuring the speed of cells traveling through narrow microfluidic channels.
- Designed and simulated a novel microfluidic ratchet chip using COMSOL Multiphysics to sort hRBCs based on age-related deformability.
- Analyzed hRBC actin-spectrin network and CD47 diffusivity on hRBC membranes using single-molecule localization microscopy (SMLM).

Immunofluorescent Biomarker for Zebrafish Somitogenesis and Single Cell Oscillation

July 2019 – Sept 2019

Summer Visiting Student | Advisor: Qiong Yang | Department of Biophysics, University of Michigan

- Developed immunofluorescent biomarkers targeting signaling pathways (*Ntla* and *Tbx16*) for studying zebrafish somitogenesis and segmentation clock dynamics.
- Identified and characterized oscillating cells with different phenotypes in zebrafish embryos and cell dispersal systems.

Conference Presentations

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- [4] "DeepSMOLM3D: Fast and Accurate 6D Single-Molecule Imaging via Deep Learning", poster. *7th Midwest Single Molecule Workshop*, University of Illinois Urbana-Champaign, Urbana, IL, July 2026
 - [3] "Combining Excitation and Emission Modulation Resolves the Angular Separation Between a Pair of Dipole Emitters", oral. *Optica Biophotonics Congress: Optics in the Life Sciences*, Coronado, CA, April 2025
 - [2] "Resolving the Orientations of and Separation between an Overlapping Pair of Dipole Emitters", poster. *Gordon Research Conferences: Single Molecule Approaches to Biology*, Newry, ME, July 2024
 - [1] "Immunofluorescent Biomarkers for Distinguishing Cell Phenotypes in Zebrafish Somitogenesis and Autonomous Cellular Oscillators", poster. *APS March Meeting 2020*, Denver, CO, March 2020 (online)

Honors and Awards

Excellence Award, China National University Student Innovation & Entrepreneurship Development Program	2021
First Prize, Nankai Physics Tournament	2018
Poling Scholarship, Nankai University	2017

Skills

Programming: Python (PyTorch, TensorFlow, NumPy/SciPy), MATLAB, Wolfram Mathematica

Software: ImageJ, COMSOL Multiphysics, Origin, PyMOL, Zemax OpticStudio

Computational Methods: inverse problems, Fisher information analysis, maximum-likelihood estimation, convex optimization, deep learning for microscopy

Laboratory

- Optical Lab: fluorescence microscopy, 4f systems, polarization-resolved imaging, SLM phase modulation, optical alignment and calibration
- Wet Lab: cell culture; immunolabeling; sample preparation

Languages: Chinese (native), English (full professional proficiency), French (elementary proficiency)

Online Licenses & Certifications:

- *Deep Learning Specialization*, DeepLearning.AI

- *Introduction to Programming with MATLAB*, Vanderbilt University

Professional Activities

Spectra, Optica & SPIE joint student chapter at *Washington University in St. Louis*

Co-president (Optica liaison)

2023 - 2024

Vice president (Imaging Science Pathway liaison)

2022 - 2023

Medical Physics Summer School at *Duke Kunshan University*

2020

The Physics of Life Summer School (virtual) at the *Center for the Physics of Biological Function*,
Princeton University

2020

Optica student chapter at *School of Physics, Nankai University*

Student officer

2018 - 2020

Teaching Experiences

Mentored students

- Cynthia Cheng (undergraduate researcher in the Lew Lab)

Fundamentals and Applications of Modern Optical Imaging

Spring 2024

Washington University in St. Louis

Assistant Instructor | Course Instructor: Matthew Lew

College Physics

Fall 2019

Nankai University

Undergraduate Teaching Assistant | Course Instructor: Jianghong Yao